

CBSE Practice Worksheet

Class 6 | Science | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
2. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
3. Which law relates induced EMF to rate of change of magnetic flux?
(A) Lenz's law
(B) Faraday's law
(C) Ampere's law
(D) Ohm's law
4. Which type of mutation involves change in single nucleotide?
(A) Deletion
(B) Insertion
(C) Point mutation
(D) Frameshift
5. What is the wavelength of de Broglie wave for an electron?
(A) $\lambda = h/mv$
(B) $\lambda = mv/h$
(C) $\lambda = hv/m$
(D) $\lambda = m/hv$
6. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
7. What is the hybridization of carbon in methane?
(A) sp
(B) sp^2
(C) sp^3
(D) sp^3d
8. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s

(D) 25 km/s

9. Which enzyme is responsible for DNA replication?

- (A) DNA ligase
- (B) DNA polymerase
- (C) Helicase
- (D) Primase

10. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?

- (A) 25%
- (B) 33%
- (C) 50%
- (D) 75%

Section B: Fill in the Blanks (1 mark each)

- 11. The band gap in semiconductors is typically ____ eV.
- 12. The enzyme that converts prothrombin to thrombin is ____.
- 13. Raoult's law applies to ____ solutions.
- 14. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 15. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 16. The uncertainty principle was given by ____.
- 17. For a reversible process, entropy change of the universe is ____.
- 18. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.

Section C: Short Answer Questions (2 marks each)

- 19. Explain the mechanism of DNA replication.
- 20. Derive the lens formula and explain its applications.
- 21. Explain the structure and working of a nuclear reactor.
- 22. Derive the equation for kinetic energy of gases from kinetic theory.
- 23. Explain Le Chatelier's principle with examples.
- 24. Describe the mechanism of nerve impulse transmission.
- 25. Explain hybridization with examples of sp , sp^2 , and sp^3 .

--- End of Worksheet ---

CBSE Practice Hub | General Practice Questions